

Yash Prakash Mishra

Backend Engineer — Node.js · NestJS · TypeScript

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Open to relocation to Berlin — on-site · English: professional fluency

PROFESSIONAL SUMMARY

Backend engineer with **5+ years in Node.js, NestJS and TypeScript**, building high-volume services where money and rewards move and mistakes are expensive. Designed and owned a **rewards and referral platform** with identity verification, **fraud prevention**, audit trails and automated programmatic payouts, lifting referral conversions **25%** and eliminating manual reward processing entirely. Also owns a **provider-agnostic payments platform across four external providers** with runtime selection, circuit-breaker fallback and idempotent, deduplicated event handling. Experienced across **MySQL, MongoDB and Redis**, Kafka and BullMQ publish/subscribe pipelines, **WebSockets** for low-latency concurrent sessions, **Jest**, Docker and CI/CD, with ELK and distributed tracing for production debugging. Leads a team of 5 and ships at startup pace; AI-assisted development (Cursor, Claude) is part of the daily workflow.

CORE SKILLS

Languages & Runtime	TypeScript, JavaScript (ES6+), Node.js, SQL, C++
Backend & APIs	NestJS, Express.js, Fastify, RESTful API design, WebSockets (Socket.io), Server-Sent Events, OpenAPI/Swagger, input validation & schema guards, JWT auth, idempotent API design, webhooks
Databases	MySQL, MongoDB, Redis, PostgreSQL, ClickHouse, Sequelize, Prisma — data modeling, zero-downtime migrations, query optimization
Messaging & Events	Apache Kafka, BullMQ, AWS SQS, publish/subscribe pipelines, event-driven architecture, event deduplication, reconciliation
External Integrations	Payment providers (Stripe, Razorpay, Affirm, Jodo), programmatic payouts (Xoxoday), identity verification, reCAPTCHA — owned end-to-end from POC to production
Testing & Quality	Jest, Supertest, Mocha, unit & integration testing, Postman, pull-request review across a 5-engineer team
Infrastructure & CI/CD	Docker, Kubernetes (working knowledge), GitHub Actions, Jenkins, AWS (EC2, S3, SQS, KMS, IAM)
Reliability & Observability	Circuit breakers (opossum), retry with backoff (axios-retry), self-healing pipelines, ELK Stack, structured logging, distributed tracing, on-call and incident reduction
Architecture	Microservices, distributed systems, multi-tenant architecture, Strategy / Factory / Singleton / Observer, HLD & LLD
AI-Assisted Development	Cursor, Claude, GitHub Copilot — implementation, code review and refactoring
Frontend (secondary)	React.js, Next.js, Redux

PROFESSIONAL EXPERIENCE

Novatr — Senior Software Engineer (Backend Developer → Senior Software Engineer) **Aug 2023 – Present**
Remote · Y Combinator-backed SaaS platform, 24/7 on AWS · Led a team of 5 engineers

Stack: NestJS, Node.js, TypeScript, PostgreSQL, MySQL, Redis, ClickHouse, Apache Kafka, BullMQ, Socket.io, Jest, Docker, AWS (EC2, S3, SQS, KMS, IAM)

Rewards, payouts & fraud prevention

- Designed and owned a **rewards and referral platform** (Node.js, TypeScript, Express.js, PostgreSQL, Kafka) combining identity verification, **fraud prevention**, audit trails and **automated programmatic payouts** via Xoxoday — driving a **25% increase in referral conversions** and eliminating manual reward processing entirely.
- Built full auditability into every reward issued: who earned it, on what basis, and which checks passed before payout — necessary because any system distributing items of cash value is an active target for abuse.

Payments & external service integration

- Architected a **provider-agnostic payments platform** integrating four external providers (Stripe, Razorpay, Affirm, Jodo) using **Strategy and Factory patterns for runtime provider selection**, so onboarding a new provider requires no changes to core checkout logic; owned each integration end-to-end from POC to production.
- Made every payment path **idempotent**, so retries, duplicate provider callbacks and client re-submissions cannot double-charge under concurrent load.
- Designed a centralised webhook handler for all four providers with self-healing retry logic, **event deduplication** and Kafka-based status reconciliation, **cutting manual error resolution by 80%**.
- Applied **circuit breakers** (opossum) and backoff retries (axios-retry) so a degraded provider fails over instead of taking the checkout path down with it.

Scalability, data & real-time — Novatr

- Introduced Kafka **publish/subscribe pipelines** (with BullMQ for in-process jobs) for payments, user lifecycle events and transactional email, decoupling services and improving fault tolerance platform-wide.
- Owned a **zero-downtime migration** of payments history off a legacy system onto a **ClickHouse**-backed analytics platform via Node.js ETL, with schema validation, integrity checks and a tested rollback path.
- Built a real-time quiz engine (Node.js, **Socket.io**, Redis) for live cohort sessions, tuned for **low-latency concurrent interactions** with graceful degradation under load.

Platform & access control

- Architected the B2C-to-B2B transition as a **multi-tenant** system (NestJS, PostgreSQL) with tenant-level data isolation, per-tenant configuration governing feature availability, and enterprise onboarding workflows — **onboarded 5+ enterprise clients** as technical point of contact.
- Architected **centralised RBAC across 5+ internal tools** with role inheritance, permission scoping and audit logging — **60% fewer access incidents**; plus a multi-layer **JWT** service (Passport.js) for token issuance, rotation and validation across three surfaces.
- Rebuilt the CMS backend (NestJS, PostgreSQL) with content lifecycle workflows and versioning — **85% faster cohort creation**, recurring on-call incidents eliminated.
- Led 5 engineers across architecture decisions, **pull-request review** and delivery timelines, partnering with product, design and QA from discovery through production.

OLX Autos — Software Engineer

May 2022 – May 2023

Remote · Used-car financing infrastructure for Latin America — regulated, high-stakes workflows

Stack: Node.js, TypeScript, Express.js, PostgreSQL, MySQL, Sequelize, REST APIs, Docker, AWS

- Built and owned foreclosure and manual repayment flows in the Loan Management System, automating loan closure and repayment tracking for **10,000+ active loans in Chile** and replacing a fully manual operations process.
- Designed **loan lifecycle state machine APIs** with validation guards at every transition — permitted states only, explicit rejections, an audit record per change — **cutting manual intervention by 40%**.
- Designed risk analysis and credit scoring workflows in the Loan Origination System, **reducing credit approval time by 30%** while improving decision auditability.
- Built a reusable PDF generation library (PDFKit, Handlebars) for programmatic invoicing in loan disbursement, **saving the operations team 10+ hours weekly**; technical point of contact for the Chile team across a 9+ hour time-zone gap.

Lenskart — Software Engineer (Frontend)

Apr 2021 – Apr 2022

India's largest eyewear retailer · Exit: project concluded on successful UAE market launch

Stack: React.js, Redux, TypeScript, Next.js, Node.js (BFF), Webpack, REST APIs

- Led the rebuild of core commerce flows (homepage, cart, checkout, order tracking) in React.js, Redux and TypeScript for the UAE market launch — **30% faster launch** with improved Lighthouse scores, as primary frontend point of contact for go-live.
- Defined response contracts against a **Node.js BFF** layer with backend teams and reduced initial JS payload through code splitting and bundle analysis — the transition point into backend engineering.

SELECTED PROJECT

BullMQ Dashboard — open-source operational tooling for distributed job pipelines

2025

Next.js (App Router), TypeScript, Node.js, Redis, ioredis, BullMQ, Express.js, Docker, SVG, Server-Sent Events · github.com/xcalibre1/bullmq-dashboard

- **Problem:** tools such as Bull Board provide queue visibility but no operational control when a pipeline misbehaves in production — no retrigger with a corrected payload, no bulk retry, no live view.
- Built a **Next.js + TypeScript operations dashboard** with **Server-Sent Events** for live queue stats, an editable JSON payload editor, a 24-hour throughput chart in pure SVG, plus payload field search, bulk retry, promote, pause and resume.
- API routes backed by a **Redis singleton** (ioredis) and a cached BullMQ Queue instance for near-zero per-request overhead; Docker Compose for one-command self-hosting. **Adopted by engineering teams globally.**

EDUCATION

B.Tech, Computer Science — Rajkiya Engineering College, Sonbhadra

Jul 2016 – Jun 2020

ADDITIONAL

- **Open source:** author and maintainer of BullMQ Dashboard (Next.js, TypeScript), used by engineering teams globally for queue operations.
- **Relocation & languages:** open to relocating to Berlin for an on-site role; professional fluency in English; 5 years working across distributed teams in India, Latin America and the Middle East.
- **Interests:** rewards and AdTech platforms, payments infrastructure, event-driven systems, developer tooling.